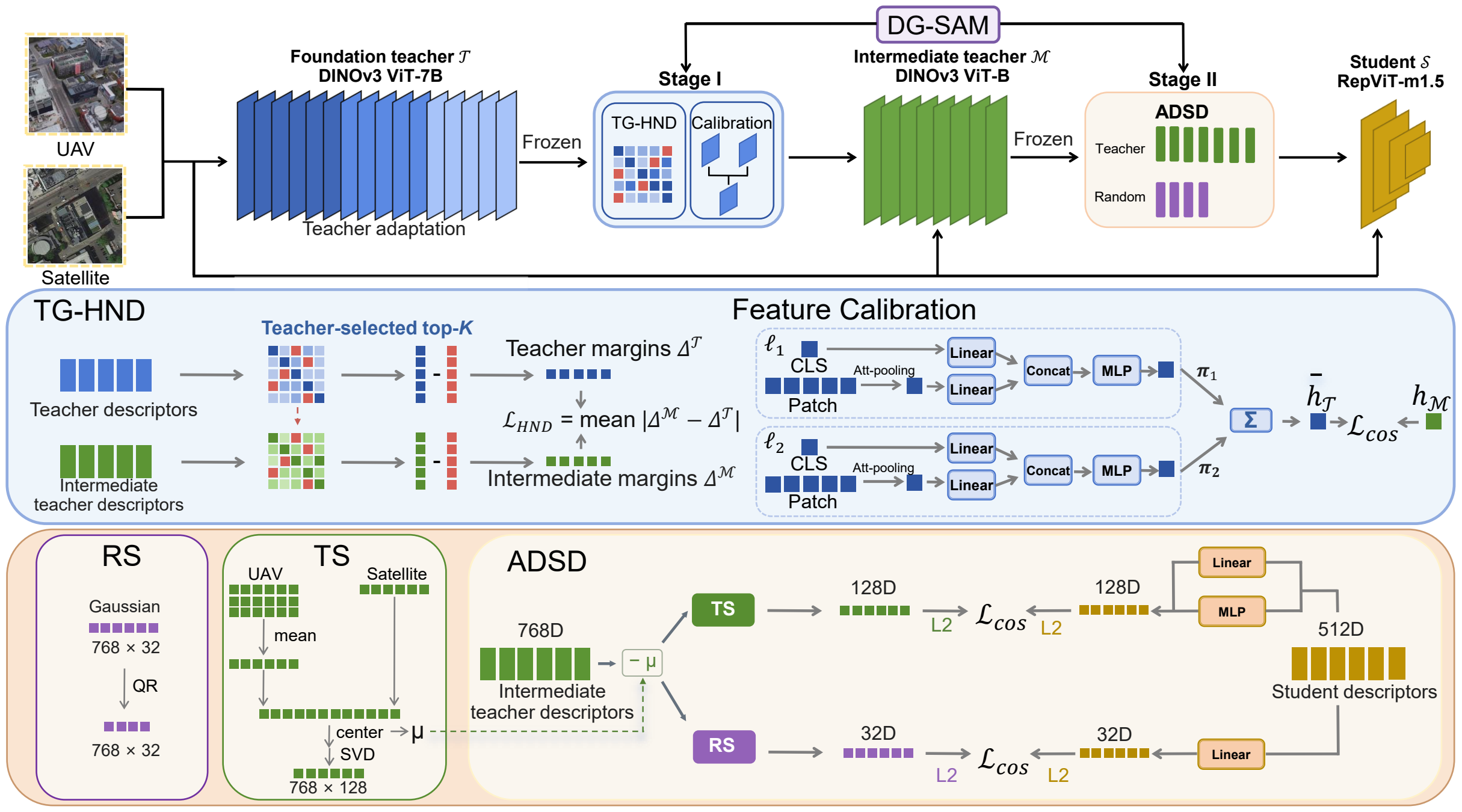

DG-SAM
UAV
Satellite
Foundation teacher $\mathcal{T}$
DINOv3 ViT-7B
Teacher adaptation
Frozen
Stage I
TG-HND
Calibration
Intermediate teacher $\mathcal{M}$
DINOv3 ViT-B
Frozen
Stage II
ADSD
Teacher
Random
Student $\mathcal{S}$
RepViT-m1.5
TG-HND
Teacher-selected top-K
Teacher descriptors
Intermediate teacher descriptors
Teacher margins $\Delta^{\mathcal{T}}$
$\mathcal{L}_{HND} = \text{mean}\,|\Delta^{\mathcal{M}} - \Delta^{\mathcal{T}}|$
Intermediate margins $\Delta^{\mathcal{M}}$
Feature Calibration
ℓ_1
CLS
Patch
Att-pooling
Linear
Linear
Concat
MLP
π_1
ℓ_2
CLS
Patch
Att-pooling
Linear
Linear
Concat
MLP
π_2
Σ
$\bar{h}_{\mathcal{T}}$
$\mathcal{L}_{cos}$
$h_{\mathcal{M}}$
RS
Gaussian
768 × 32
QR
768 × 32
TS
UAV
Satellite
mean
center
μ
SVD
768 × 128
ADSD
768D
Intermediate teacher descriptors
− μ
TS
RS
128D
L2
$\mathcal{L}_{cos}$
L2
128D
Linear
MLP
512D
Student descriptors
32D
L2
$\mathcal{L}_{cos}$
L2
32D
Linear